Design and Implementation of a Intelligent Virtual Agent for Kabré Supercomputer Infrastructure Support

Dylan Benavides[1,2][0009−0006−8777−7876]

[1] National High Technology Center, San José 10109, Costa Rica
dbenavides@cenat.ac.cr
https://www.cenat.ac.cr/es/
[2] University of Costa Rica, San José 11501-2060, Costa Rica
dylan.benavides@ucr.ac.cr
https://www.ucr.ac.cr

Abstract. abstract

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1 Problem Definition

High-Performance Computing (HPC) environments typically host highly heterogeneous user communities composed of students, researchers, and professionals from diverse disciplines. This diversity naturally leads to a wide range of technical expertise, resulting in queries of varying complexity—from basic questions about system usage and command-line operations to advanced issues involving software installation, resource optimization, and memory management.

In Costa Rica, the Kabré supercomputer, operated by the National High Technology Center (CeNAT), represents a critical infrastructure for scientific research. The system currently supports approximately 300 active users, generating over 100,000 computing hours annually. This community is primarily composed of students, faculty members, and researchers from public universities across the country, spanning multiple domains of knowledge. Despite its importance, user support poses a significant operational challenge, as it is managed by a small team consisting of a single infrastructure engineer and one technical assistant.

At present, user support is handled through a ticket-based system, where users submit queries that may include problem descriptions, code snippets, and supporting files. While functional, this approach presents several limitations. First, it introduces friction in the user experience, as users must navigate the platform, authenticate, and formally structure their requests before receiving assistance. Second, resolving issues often requires multiple back-and-forth interactions to clarify context, which increases response and resolution times.

Furthermore, the system becomes overloaded due to both the volume of requests and the recurrence of similar queries. Many of these correspond to frequently asked questions that require repetitive responses, leading to inefficient

use of the support team's time. At the same time, more specialized queries—such as those arising in fields like bioinformatics or computational sciences—require in-depth, domain-specific analysis. This combination of routine and complex queries contributes to system inefficiency and the accumulation of unresolved tickets.

Although a knowledge base exists within the current system, it is outdated, limited in scope, and lacks interactivity. Users cannot perform targeted queries or obtain immediate answers, forcing them to rely entirely on the ticketing system. As a result, response times can exceed 24 hours, which is particularly problematic in scenarios requiring rapid troubleshooting or near real-time assistance.

Given these challenges, there is a clear need to leverage modern technological solutions to improve user support in HPC environments. In particular, the development of an intelligent virtual agent emerges as a promising approach to automate the handling of user queries, especially those that are repetitive in nature. Such a system could provide real-time assistance, reduce the operational burden on infrastructure staff, and enhance the overall user experience. However, in its initial stages, the agent should not operate with full autonomy; instead, it should work in conjunction with human experts, determining which queries can be effectively resolved automatically and which require escalation to the support team.

2 Literature Review

Early research in High-Performance Computing (HPC) began to challenge the notion that computational performance alone defines system effectiveness, instead emphasizing the importance of interaction, usability, and human-centered design. One of the earliest contributions in this direction is the work of Faigle, Fox, Furmanski, Niemiec, and Simoni (1993), who established that virtual reality (VR) and HPC are inherently complementary technologies. They argued that large-scale "Grand Challenge" simulations require dynamic VR interfaces capable of visualizing complex computational processes in real time. A key technical insight from their work is the need to decouple the update rates of HPC computational engines from VR display systems to prevent instability and user discomfort, such as simulation sickness [7]. Despite these advances, the authors identified critical limitations, including the lack of standardized connectivity models and the absence of hardware-independent interface protocols. These challenges remain relevant today, highlighting a persistent gap between computational performance and interactive usability that modern HPC systems continue to address [7].

Building on the importance of interaction, Cai and He (2004) introduced an early approach to enhancing human–machine communication through avatar-based interfaces. At the time, most interaction systems relied on rigid rule-based mechanisms and static learning models that were unable to adapt to different users or contexts. To overcome these limitations, the authors proposed the Interactive Agent Based on Awareness (ABI Agent), a system designed to simulate

human-like behavior through a structured process of intention generation, goal selection, and action planning. Their results demonstrated that embedding behavioral characteristics into agents can significantly improve user efficiency by automating repetitive tasks and enabling more intuitive interaction.

However, Cai and He (2004) also acknowledged fundamental challenges in achieving truly adaptive and human-like systems. As discussed in [24], integrating personality and human knowledge into intelligent agents remains difficult due to the complexity and variability of psychological and contextual factors across users. Additionally, early systems lacked the ability to learn autonomously from interaction, limiting their capacity to adapt dynamically to user needs. These limitations underscore the broader challenge of moving from static, rule-based systems toward more flexible, learning-driven interaction models.

A more explicit focus on the human dimension in HPC emerges in the work of Monroy, Becerra, Bellas, and Duro (2006, 2008), who argue that traditional optimization strategies are often misaligned with user needs. Rather than prioritizing metrics such as job turnaround time or queue waiting time, they emphasize that the ultimate objective of HPC systems should be to maximize user satisfaction while minimizing inefficiencies. Their studies reveal that users frequently overestimate required computational resources—particularly CPU time—as a precaution against job failure, leading to suboptimal scheduling and reduced system efficiency.

To address this issue, the authors developed the EVOPROC interface, which combines neural networks and evolutionary algorithms to predict actual resource consumption more accurately than manual user submissions [1,**?**]. This approach enables HPC systems to interpret imprecise user inputs and translate them into optimized execution plans, effectively bridging the gap between human intent and system-level performance. Furthermore, they identify behavioral indicators such as frequent system queries, including `qstat`, as signals of user uncertainty and impatience, highlighting the importance of incorporating behavioral data into system design.

Subsequent research, including [10], reinforces the importance of user behavior modeling as a critical yet underdeveloped component of HPC scheduling systems. In particular, these studies point out the difficulty of formalizing subjective dimensions such as satisfaction, trust, and perceived responsiveness, which vary significantly across users and contexts. As a result, a persistent gap remains between technical performance metrics and human-centered evaluation.

Moreover, both Monroy et al. and later studies highlight a structural limitation in traditional HPC schedulers: although they provide extensive configuration parameters, there is often no clear mapping between these parameters and the quality of service perceived by users. This disconnect is further exacerbated by users' limited understanding of their own computational requirements. Consequently, there is a growing need for intelligent systems capable of translating uncertain or conservative user inputs into efficient and reliable execution strategies.

Overall, these foundational works mark a critical transition in HPC research—from systems focused purely on computational optimization toward environments that increasingly account for human interaction, perception, and adaptability. While early approaches laid the groundwork for integrating usability into HPC, they also revealed significant limitations that continue to motivate the development of more intelligent, user-aware, and autonomous systems.

To bridge the early human-centered vision of HPC with more mature intelligent interaction systems, recent research begins to formalize how user experience, personalization, and adaptive assistance can be operationalized at scale. While foundational works established the importance of visualization, user satisfaction, and basic agent behavior, the next stage of research shifts toward embedding intelligence directly into support systems, enabling them to interpret user needs, learn from interaction data, and provide context-aware assistance. This transition marks the emergence of a second layer in HPC evolution, where interaction is no longer passive but actively mediated by intelligent, adaptive systems.

Within this context, Biggs and Dalton (2024) highlight a persistent knowledge gap among researchers, emphasizing that expertise in one HPC system does not easily transfer to another. To address this, they demonstrate that integrating AI assistants based on *RAG* enables highly precise, site-specific technical support grounded in continuously updated internal documentation, thus avoiding the costly need for frequent model retraining [5]. However, they also identify a critical limitation: historical support data—such as tickets—often contain noise, including irrelevant conversations or sensitive information, which complicates their transformation into reliable knowledge sources.

This challenge of creating more natural and adaptive interaction systems echoes earlier concerns raised by Cai and He (2004), who argue that traditional avatar-based interfaces lack personality and flexibility. Their proposed ABI Agent represents an important step toward more human-like interaction, introducing a structured process of intention generation, goal selection, and action planning. By embedding behavioral tendencies into agents, their approach demonstrates measurable improvements in user efficiency, particularly in handling repetitive or cognitively demanding tasks. Nevertheless, they acknowledge that fully modeling human personality and contextual understanding remains a complex and unresolved problem, especially given the variability of psychological and sensory factors across users [24].

Expanding on the limitations of interaction design, Diederich et al. (2020) provide a broader perspective on conversational agents, arguing that many systems fail not because of technical shortcomings, but due to their neglect of user-specific characteristics such as culture, personality, and expectations. They advocate for longitudinal, real-world studies to better understand how social and behavioral cues influence long-term user engagement and system sustainability [22]. This reinforces the idea that effective HPC support systems must go beyond functional correctness and incorporate deeper human-centered design principles.

From an operational standpoint, Sanjeewa (2020) situates these challenges within the broader landscape of technical support systems, noting that the grow-

ing complexity of enterprise environments places increasing pressure on support infrastructures. The study emphasizes that AI-driven support tools are most effective when deployed in a human-in-the-loop configuration, where chatbots handle repetitive queries while experts focus on complex issues. However, important gaps remain, including the lack of specialized ticketing systems and the need for continuous learning mechanisms driven by real user feedback, as well as enhanced security features such as multi-factor authentication.

Building on these ideas within the HPC domain, Shifare, Shukla, and Kothari (2025) demonstrate how intelligent systems can significantly improve both access management and user support. Their AI-based Laboratory Manager (ALM) reduces resource access approval times from days to minutes, while a fine-tuned RoBERTa-based chatbot achieves a 98.1% accuracy rate in resolving user queries. Despite these advances, the study highlights persistent issues such as ambiguity in user requests and inconsistencies in documentation, while also pointing toward future directions in predictive resource management and more advanced scheduling optimization [13].

Finally, Wulf and Meierhofer (2024) extend this discussion by analyzing the role of large language models in technical customer service environments. They show that while LLMs are highly effective for low-level cognitive tasks such as summarization and translation, more complex problem-solving requires enhanced architectures like *RAG* or fine-tuning to reduce hallucinations. Their work also underscores the importance of structured data ecosystems to enable context-aware AI systems. However, they identify a critical gap in the lack of large-scale, real-world validation studies, as well as challenges related to user acceptance, usability, and integration with legacy systems [21].

Together, these studies define Cluster 2 as a transition toward intelligent, user-centered support systems in HPC, where the focus shifts from static interfaces to adaptive, learning-driven interaction layers capable of bridging the gap between complex computational infrastructures and diverse user communities.

As intelligent interaction layers mature and begin to effectively mediate between users and complex HPC infrastructures, the next logical step is to optimize what happens behind the interface: the allocation, scheduling, and orchestration of computational resources. This shift marks a transition from user-centered assistance toward system-level intelligence, where AI is not only supporting users but actively managing workloads, predicting behavior, and optimizing performance in real time. In this sense, Cluster 3 establishes the operational backbone upon which higher-level automation—such as fully autonomous support systems—can be built.

Within this framework, Fan et al. (2020) demonstrate that traditional scheduling heuristics like FCFS are fundamentally inadequate for dynamic HPC environments, as they fail to adapt to fluctuating workloads and often require continuous manual tuning. Their proposed DRAS system leverages hierarchical deep reinforcement learning to autonomously manage resource reservation and backfilling, achieving performance improvements of up to 45%. However, the authors highlight a critical limitation: models originally designed for cloud computing

do not translate well to HPC due to factors such as job rigidity and starvation, revealing an important gap in domain-specific AI design [15].

Complementing this work, Narantuya et al. (2022) expand the optimization perspective by incorporating energy efficiency into the scheduling problem. Their decentralized multi-agent deep reinforcement learning ($mDRL$) approach significantly improves both performance and sustainability, reducing task completion times by 20% and inefficient energy consumption by 40% [16]. Despite these gains, the study exposes structural limitations in current HPC ecosystems, particularly regarding accessibility for smaller institutions and the inability of centralized architectures to meet low-latency demands. This reinforces the need for hybrid models that integrate Edge Computing with traditional HPC infrastructures.

From a behavioral standpoint, Zaborovsky et al. (2023) reveal that inefficiencies in HPC systems are not only technical but also human-driven. Their analysis shows that users overestimate job execution times in more than 89% of cases, leading to poor resource utilization and congested queues. To address this, they propose a multi-agent scheduling system enhanced with Random Survival Forests (RSF), which predicts probabilistic execution times rather than fixed estimates. While this improves accuracy, the authors emphasize the necessity of Explainable AI (XAI) to ensure that users understand system decisions, and they advocate for the integration of Natural Language Processing (NLP) interfaces to facilitate more intuitive interaction with scheduling systems [14].

In parallel, Jain, Gupta, and Neha (2024) tackle inefficiencies from the perspective of support infrastructure, arguing that traditional ticket management systems are ill-equipped to handle the complexity of modern IT environments. Their multidimensional clustering approach—combining semantic, spatial, and temporal features—achieves a fidelity of 96% while reducing manual workload by 30% [18]. However, they identify a major barrier in the transition from legacy systems to AI-driven frameworks, particularly in preserving institutional knowledge and overcoming organizational resistance.

At the architectural level, Oparin, Bogdanova, and Pashinin (2020) introduce the $HPCSOMAS$ framework, which integrates hierarchical control with decentralized peer-to-peer agent networks. This hybrid approach enables high scalability and flexibility in solving complex distributed problems, yet it also highlights the growing complexity of such systems, especially for non-expert users. The lack of high-level abstraction tools remains a key obstacle for broader adoption [25].

Finally, Rosendo et al. (2025) push the boundaries of automation by proposing AI-driven orchestration of scientific workflows through modular agent-based architectures. By combining natural language interfaces with programmable APIs, their system enables adaptive and near real-time coordination of computational resources, significantly accelerating scientific experimentation [4]. Nonetheless, they identify unresolved challenges related to cross-facility integration, as well as limitations imposed by existing security policies that are not designed for interactive or streaming workflows.

Together, these contributions define Cluster 3 as the core of AI-driven optimization in HPC, where scheduling, resource management, and workflow orchestration converge into intelligent, adaptive systems.

Building upon this operational intelligence, the next stage of evolution—Cluster 4—shifts the focus toward closing the loop between system optimization and user support. While Cluster 3 optimizes how resources are allocated and managed, Cluster 4 explores how large language models, retrieval systems, and conversational agents can make these complex systems accessible, interpretable, and increasingly autonomous from the user's perspective.

Building upon the advances in AI-driven scheduling and resource optimization, the final stage in this evolution focuses on closing the loop between system intelligence and user interaction. While Cluster 3 demonstrates how HPC infrastructures can autonomously manage workloads and optimize performance, Cluster 4 shifts attention toward making these capabilities accessible, interpretable, and actionable for users. In this layer, large language models (LLMs), retrieval-augmented generation (RAG), and conversational agents emerge as the key enablers of fully integrated, intelligent support ecosystems—where interaction, automation, and decision-making converge into a unified experience.

At the core of this transition, Rosendo et al. (2025) propose a modular, agent-based architecture that bridges workflow orchestration with natural language interfaces and programmable APIs. Their approach enables adaptive and explainable coordination of scientific workflows in near real time, effectively reducing the reliance on manual intervention [4]. However, they identify persistent barriers in cross-facility integration, particularly due to misaligned security policies and the lack of standards for interactive or streaming workloads—highlighting a structural limitation in current HPC ecosystems.

Complementing this systems-level perspective, Wulf and Meierhofer (2024) explore the role of LLMs in automating technical support tasks. They demonstrate that while models such as GPT-4 are highly effective for low-level cognitive operations (e.g., summarization and translation), more complex reasoning tasks require enhanced architectures like RAG or fine-tuning to mitigate hallucinations. Their work underscores the importance of structured data ecosystems for enabling context-aware AI, yet also reveals a critical gap in large-scale, real-world validation and system integration [21].

Addressing the challenge of fragmented knowledge, Bondapalli et al. (2025) show that HPC documentation—often distributed across heterogeneous formats—can be effectively leveraged through multimodal RAG systems. By preserving the structural integrity of technical content such as code and tables, their approach significantly improves response relevance, achieving a score of 4.7/5.0 [20]. Nevertheless, they identify that retrieval quality remains highly sensitive to noisy or irrelevant context, motivating the need for more advanced techniques such as hybrid search and knowledge graph integration.

From an applied support perspective, Pasupuleti et al. (2025) demonstrate the practical impact of LLM-based chatbots in HPC environments. Their sys-

tem, enhanced with FAISS and BERT-based semantic retrieval, is capable of autonomously resolving up to 70

Similarly, Shifare et al. (2025) present an integrated solution combining an AI-based Laboratory Manager (ALM) with a RoBERTa-based chatbot, achieving a 98.1

Extending beyond support automation, Yin et al. (2024) introduce the chatHPC architecture, demonstrating how LLMs can be fully integrated into HPC infrastructures to enable programmable and conversational scientific workflows. Their results show significant improvements in model training efficiency and response accuracy through a hybrid combination of supervised fine-tuning and RAG. However, they point out unresolved challenges in scalable model optimization, dataset generation, and hallucination mitigation, as well as the need for tighter integration with simulation frameworks for end-to-end automation [19].

From a communication standpoint, Guo et al. (2021) contribute to the development of more natural interaction modalities by enhancing text-to-speech systems with contextual and semantic encoders. Their approach enables more human-like conversational agents capable of capturing spontaneous speech patterns, which is essential for immersive and effective user support [2]. Complementarily, Ashish et al. (2025) expand the role of AI agents into the domain of human collaboration, demonstrating how virtual mentors powered by lightweight LLMs can facilitate intercultural communication in global HPC projects, although challenges remain in long-term evaluation and curricular integration [6].

Finally, Jain et al. (2024) reinforce the importance of intelligent support infrastructures by showing how multidimensional AI models can significantly improve incident management through more accurate ticket clustering and resolution recommendations. Despite achieving high performance, their work highlights ongoing challenges in adaptability, contextual reasoning, and the transition from legacy systems—issues that directly impact the deployment of LLM-based support ecosystems at scale [17].

Taken together, these contributions define Cluster 4 as the most advanced and integrative layer in the evolution of HPC systems. Here, intelligent agents, LLMs, and retrieval-based architectures converge to create adaptive, context-aware, and increasingly autonomous support ecosystems. Rather than simply assisting users or optimizing resources in isolation, these systems aim to unify interaction, computation, and knowledge into a cohesive framework capable of supporting the full lifecycle of scientific discovery.

3 Research Questions

Based on the challenges identified in the Kabré HPC support system, this study explores the design, implementation, and impact of an Intelligent Virtual Agent (IVA). The research is guided by the following main questions:

1. To what extent can an Intelligent Virtual Agent improve the efficiency of technical support in the Kabré supercomputer environment, particularly in terms of response time, resolution time, and overall system performance?
2. To what extent can the introduction of an IVA reduce the workload of the technical support team by handling repetitive and frequently asked queries?

In addition to these main questions, the study also considers several complementary aspects. These include the role of multimodal features such as blendshapes for lip-sync and whether they provide meaningful value in user interaction or remain largely irrelevant in an HPC support context. It also examines the impact of the IVA on user satisfaction, the types of data that can be used to train the underlying language model, and how data quality influences system performance. Furthermore, the study explores how autonomy can be designed so that the agent can decide which queries to resolve automatically and which to escalate to human experts. Security considerations are also addressed, particularly in scenarios where the IVA is granted elevated permissions, requiring safeguards to prevent risks to the HPC infrastructure. Finally, the study evaluates which types of queries can be reliably handled by the IVA, based on validation with domain experts, as well as the system's ability to communicate in a clear and empathetic manner across a diverse user base.

4 Objectives of the Agent

The Intelligent Virtual Agent (IVA) is designed to act as a first-line support system for the Kabré High-Performance Computing (HPC) infrastructure. Its main objectives are:

1. To provide real-time assistance to users by resolving common technical issues related to system usage, job execution, software installation, and resource management.
2. To reduce the workload of the human support team by handling repetitive and frequently asked questions, while ensuring that complex cases are escalated when necessary.
3. To improve the overall user experience within the HPC environment by offering clear, adaptive, and accessible guidance tailored to different user profiles.

5 Preliminary Technical Architecture

The architecture of the Intelligent Virtual Agent (IVA) is designed as an integrated system combining large language models, voice services, and an interactive interface to support the technical assistance process within the Kabré HPC environment. The design is motivated by the heterogeneity of the user community, which implies that interactions cannot follow a single uniform behavior. Instead, the agent is expected to adapt its responses according to predefined

user clusters such as social sciences, natural sciences, or computational disciplines, an approach consistent with recent work on LLM-based support systems in HPC environments [21] and findings in conversational agent research highlighting the importance of user characteristics and adaptive interaction design in socio-technical systems [22]. This adaptation affects both the level of technical detail and the explanatory style used in responses.

At the core of the system, the Gemma 4 E4B model is used, deployed within the Kabré infrastructure. This model is selected due to its ability to operate efficiently in high-performance computing environments and its strong performance in natural language understanding and generation tasks. These characteristics make it suitable for handling a wide range of technical support queries, from basic system usage to more complex issues such as job execution, software installation, and resource management.

Interaction with the system is based exclusively on textual input from the user. Users communicate with the agent only through text, without any speech input capability. However, the system incorporates VoxCPM2 as a text-to-speech (TTS) engine for the agent's output. In this configuration, the language model generates textual responses that are subsequently converted into speech, enabling a multimodal output experience while maintaining a simple and consistent text-only input interface. The use of speech synthesis in this context is motivated by advances in conversational neural TTS systems, which show that incorporating conversational context and semantic awareness significantly improves prosody naturalness and perceived realism in voice agents [23]. Such approaches demonstrate that high-quality conversational TTS benefits from modeling dialogue context and utterance-level dependencies, leading to more natural and contextually appropriate speech generation.

The user interface is developed using Unity, enabling a real-time interactive environment for communication with the agent. This interface serves as the main point of interaction between users and the support system, complementing the traditional ticket-based workflow. The use of an avatar-based representation of the agent is intended to make the interaction more intuitive and accessible, particularly for less technical users. This design choice is consistent with research on interactive agent-based avatar systems, where embodied virtual agents have been shown to improve usability and engagement by combining visual representation with structured behavioral models [24].

Additionally, the system may incorporate facial animation technologies such as the NVIDIA LipSync model to improve synchronization between generated speech and visual representation. This component is considered optional and is intended to enhance the naturalness of the interaction experience.

The system is designed to run primarily within the Kabré infrastructure, leveraging its computational resources for both the language model and backend services. The IVA will be accessible exclusively through the official Kabré website, operating as an online service integrated into the institutional portal, where users can interact with the agent directly from their web browser.

Overall, this architecture is intended to address the main limitations identified in the current support system, including ticket overload, user heterogeneity, and long response times. Rather than replacing human support, the system is designed to complement it by automating repetitive queries while escalating complex cases to human experts when necessary.

6 GitHub Repository

The GitHub repository contains the project structure and documentation of the intelligent virtual agent, including the design proposal and supporting materials for its implementation in HPC support for the Kabré environment. It also includes the initial project organization, with space reserved for the future integration of code components.

https://github.com/DylanBC09/Intelligent-Virtual-Agent-for-Kabr-Supercomputer-Infrastructure-Support-/blob/main/README.md

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